ASSIGNMENT-8.5

Name: Mohd Manzoor Ahmed

Hall No:2303A53042

Batch: 46

Task Description #1 (Username Validator – Apply AI in Authentication Context)

- Task: Use AI to generate at least 3 assert test cases for a function is_valid_username(username) and then implement the function using Test-Driven Development principles.
- Requirements:

o Username length must be between 5 and 15 characters. o Must contain only alphabets and digits. o Must not start with a digit. o No spaces allowed.

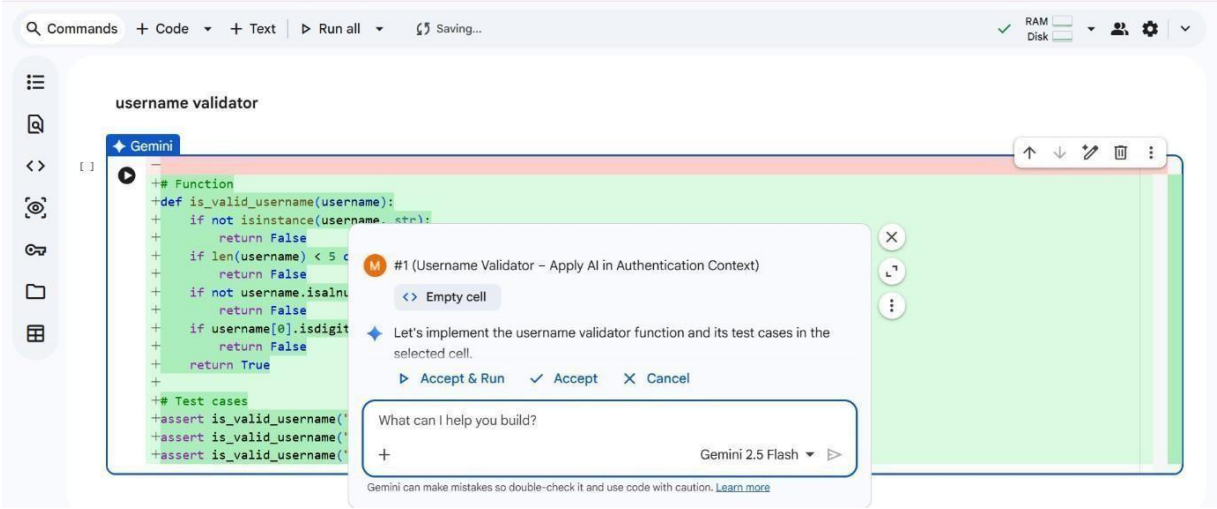
Example Assert Test Cases:

```
assert is_valid_username("User123") == True
assert is_valid_username("12User") == False
assert is_valid_username("Us er") == False
```

Expected Output #1:

- Username validation logic successfully passing all AI-generated test cases.

Output:



Task Description #2 (Even–Odd & Type Classification – Apply AI for Robust Input Handling)

- Task: Use AI to generate at least 3 assert test cases for a function classify_value(x) and implement it using conditional logic and loops.
- Requirements:

o If input is an integer, classify as "Even" or "Odd". o If input is 0, return "Zero".

o If input is non-numeric, return "Invalid Input". Example Assert Test Cases: assert classify_value(8) == "Even" assert classify_value(7) == "Odd" assert classify_value("abc") == "Invalid Input" Expected Output #2:

- Function correctly classifying values and passing all test cases.

Output:

Task Description #3 (Palindrome Checker – Apply AI for String Normalization)

- Task: Use AI to generate at least 3 assert test cases for a function is_palindrome(text) and implement the function.
- Requirements:

o Ignore case, spaces, and punctuation.

o Handle edge cases such as empty strings and single characters.

Example Assert Test Cases:

assert is_palindrome("Madam") == True

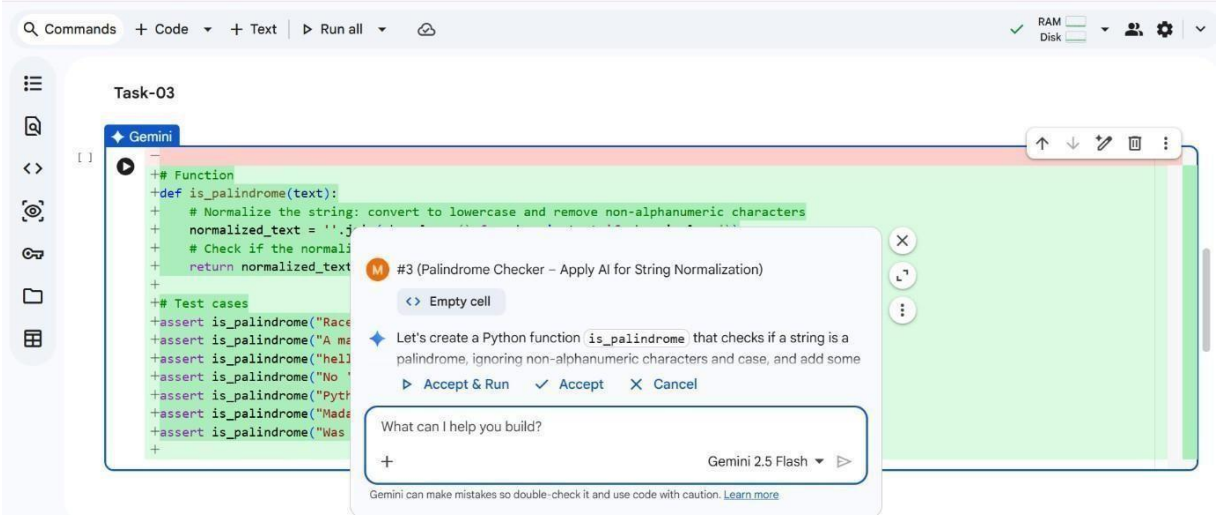
assert is_palindrome("A man a plan a canal Panama") == True

assert is_palindrome("Python") == False Expected Output

#3:

- Function correctly identifying palindromes and passing all AI-generated tests.

Output:



Task Description #4 (BankAccount Class – Apply AI for Object-Oriented Test-Driven Development)

- Task: Ask AI to generate at least 3 assert-based test cases for a BankAccount class and then implement the class.

- Methods:

o deposit(amount) o

withdraw(amount) o get_balance()

Example Assert Test Cases: acc =

BankAccount(1000) acc.deposit(500)

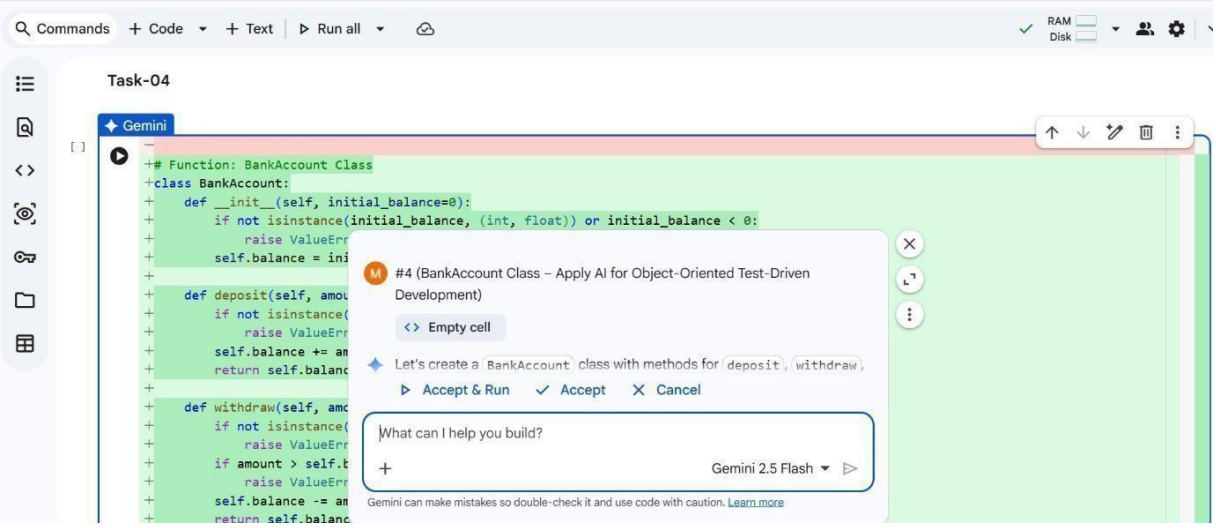
assert acc.get_balance() == 1500 acc.withdraw(300) assert

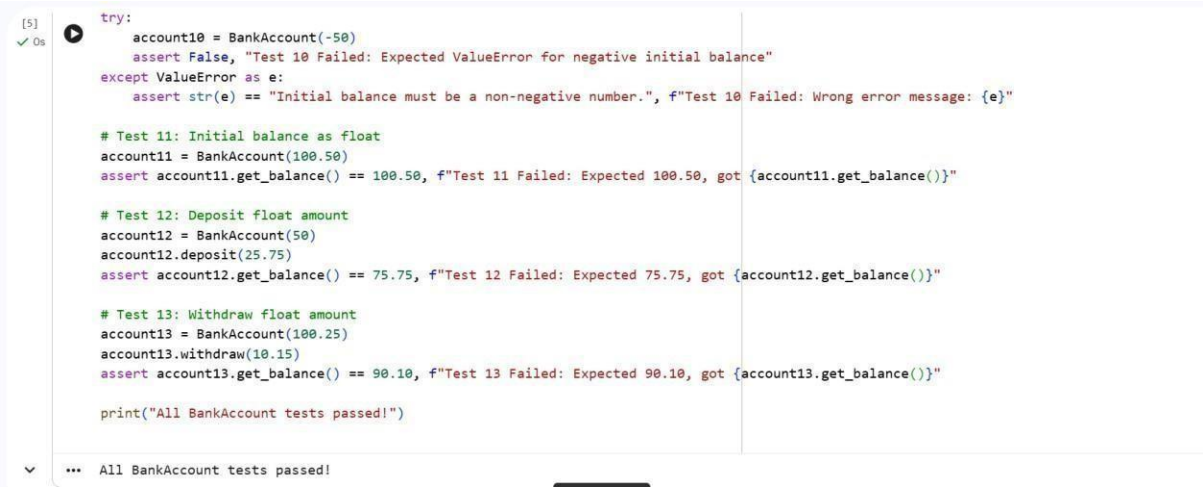
acc.get_balance() == 1200 Expected Output

#4:

- Fully functional class that passes all AIgenerated assertions.

Output:





```
try:
    account10 = BankAccount(-50)
    assert False, "Test 10 Failed: Expected ValueError for negative initial balance"
except ValueError as e:
    assert str(e) == "Initial balance must be a non-negative number.", f"Test 10 Failed: Wrong error message: {e}"

# Test 11: Initial balance as float
account11 = BankAccount(100.50)
assert account11.get_balance() == 100.50, f"Test 11 Failed: Expected 100.50, got {account11.get_balance()}"

# Test 12: Deposit float amount
account12 = BankAccount(50)
account12.deposit(25.75)
assert account12.get_balance() == 75.75, f"Test 12 Failed: Expected 75.75, got {account12.get_balance()}"

# Test 13: Withdraw float amount
account13 = BankAccount(100.25)
account13.withdraw(10.15)
assert account13.get_balance() == 90.10, f"Test 13 Failed: Expected 90.10, got {account13.get_balance()}"

print("All BankAccount tests passed!")
```

```
All BankAccount tests passed!
```

Task Description #5 (Email ID Validation – Apply AI for Data Validation)

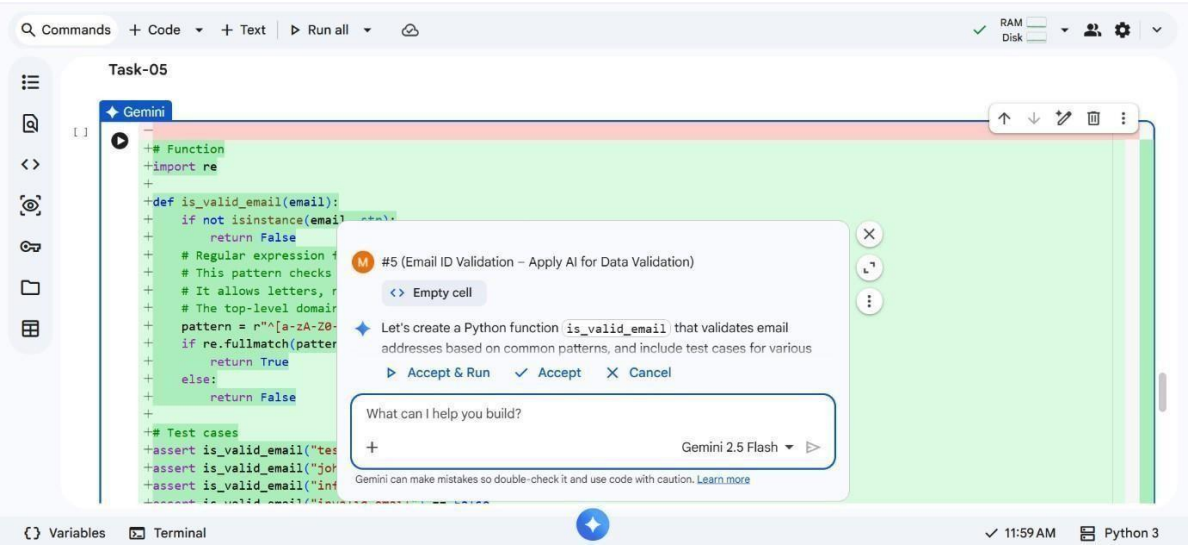
- Task: Use AI to generate at least 3 assert test cases for a function validate_email(email) and implement the function.

- Requirements:

o Must contain @ and . o Must not start or end with special characters. o Should handle invalid formats gracefully. Example Assert Test Cases:

assert validate_email("user@example.com") == True assert validate_email("userexample.com") == False assert validate_email("@gmail.com") == False

Expected Output #5:

- Email validation function passing all AI-generated test cases and handling edge cases correctly.

Output: